Technical methods behind my Bayesian DAG

The arrows stay fixed. These methods learn their strengths and turn inputs into a forecast.

METHOD	WHAT I USED	WHY I USED IT
Fixed DAG + linear Gaussian node models	Each value = weighted inputs + Gaussian noise	Explicit assumptions; keep percentage scores continuous
Normal–Inverse-Gamma shrinkage	Bayesian weight / noise estimates Coefficient precisions: 1 vs 10	Limit unstable weights from small samples / overlapping marks
Condition the joint Gaussian on observed inputs	Compute P(final % | inputs) from the fitted joint model	Personalize the forecast using what is known for that student
Marginalize missing evidence	Integrate over unknown inputs at prediction time	Predict when some student inputs are not reported
Final-% mean + 90% interval	Mean μ; Gaussian μ ± 1.645σ Optional past-error rescaling	Show the expected result and an uncertainty range

Training gaps use medians. Main intervals use fitted parameters; they do not include full parameter uncertainty.